

RADIONUCLIDE SAFETY DATA SHEET																							
RADIONUCLIDE: Tc-99			FORMS: Soluble																				
PHYSICAL CHARACTERISTICS HALF-LIFE: 211,100 years DECAY EMISSIONS																							
<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="2" style="text-align: left; padding: 2px;">Gammas / X-rays</th> <th colspan="2" style="text-align: left; padding: 2px;">Betas / Positrons (+) / Electrons*</th> <th colspan="2" style="text-align: left; padding: 2px;">Alphas</th> </tr> <tr> <th style="text-align: left; padding: 2px;">E (keV)</th> <th style="text-align: left; padding: 2px;">%</th> <th style="text-align: left; padding: 2px;">E (keV, Ave)</th> <th style="text-align: left; padding: 2px;">%</th> <th style="text-align: left; padding: 2px;">E (keV)</th> <th style="text-align: left; padding: 2px;">%</th> </tr> <tr> <td style="height: 40px;"></td> <td></td> <td>101</td> <td>100</td> <td></td> <td></td> </tr> </table>		Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas		E (keV)	%	E (keV, Ave)	%	E (keV)	%			101	100						
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E (keV)	%	E (keV, Ave)	%	E (keV)	%																		
		101	100																				
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.																							
STANFORD HAZARD CATEGORY C – level (low hazard): ≤ 20 mCi B – level (moderate hazard): > 20 mCi, ≤ 1 Ci A – level (high hazard): > 1 Ci																							
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: ~0 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 19 mrem/h Contamination skin dose, uniform deposit of 1 µCi per cm²: 4300 mrem/h			INTERNAL RADIATION HAZARDS Annual Limit on Intake: 4000 µCi (Ingestion) 700 µCi (Inhalation) The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).																				
SHIELDING Gammas/X-rays: N/A Betas/electrons: 0.6 mm of plastic will absorb all emissions. Bremsstrahlung may be created and require additional shielding.			DOSIMETRY AND BIOASSAY REQS Whole-body and finger-ring dosimeters are required for handling 5 mCi or more, or 1 mCi amounts weekly . Urine assays may be required after large spills or contaminations.																				
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: PGM 2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 3. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required. 4. Segregate waste to those with half-lives greater than 120 days (excluding H3 and C14). Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot. 5. Limit soluble waste to the sewer to less than 100 µCi/day per lab.																							

References:

- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2nd ed.). Ashford, Kent: Nuclear Technology Publishing.
- Johnson, T.E., Birkby, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>